

Aditya Kathe

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EDUCATION

SVKM's Dwarkadas J. Sanghvi College of Engineering

Bachelor of Technology in Computer Engineering, Honors in Data Science (CGPA: **8.67**)

Mumbai, India

2023 – 2027

TECHNICAL SKILLS

Languages: Java, Python, SQL, Dart

AI & Agentic Systems: LLMs, LangGraph, LangChain, RAG, pgvector, Prompt Engineering, AI Agents, GitHub Copilot

Backend & APIs: FastAPI, Spring Boot, REST APIs, WebSockets, JPA/Hibernate, RabbitMQ

Databases & Analytics: PostgreSQL, MySQL, MongoDB, Redis, Power BI

Infrastructure & Tools: Docker, Git, GitHub, Postman, Firebase, Antigravity, Codex, Cursor, Claude, Microsoft Copilot Studio

EXPERIENCE

Software Intern

Apr 2026 – Present

Triponovaa | dev.triponovaa.com

- Built an AI-powered travel assistant using **LangGraph** and an internally hosted open-source LLM, orchestrating flight, hotel, and travel-insurance workflows through natural-language intent, tool calling, and external APIs.
- Implemented **stateful, iterative agent workflows** with guardrails for request clarification and rejection, dynamically invoking search and user-information tools while carrying context across search, booking, and confirmation stages.
- Developed **verification-agent** steps that compared API results and booking details against the user's original requirements, enabling validation and retry handling before progressing through the workflow.
- Integrated post-booking automation for receipt generation, email delivery to primary and additional recipients, and in-app notifications, while routing payment through a secure embedded **WebView** rather than exposing payment execution to the agent.

Application Developer

Mar 2025 – Apr 2026

SVKM EduConnect | educonnect.svkm.ac.in

- Built and shipped a cross-platform school communication platform with **1000+** downloads across Play Store and App Store, supporting **40+** teacher profiles with expansion ongoing.
- Automated fragmented communication and administrative workflows by replacing WhatsApp groups, paper circulars, and manual attendance with unified announcements, leave requests, appointments, and messaging.
- Implemented real-time messaging with WebSockets and workflow logic across attendance, timetable, leave, and appointment modules while collaborating directly with stakeholders on product iterations.

PROJECTS

Distributed Job Scheduler | aditya-10k/job-scheduler | Java, Spring Boot, PostgreSQL, Apache Kafka, Docker Aug 2026

- Built a distributed job scheduling platform with Sprg KaKafka to support scheduled job creation, persistent state management, and horizontally scalable scheduler and worker services.
- Implemented atomic job claiming with conditional SQL updates and a transactional outbox pattern, preventing duplicate claims across concurrent scheduler instances and ensuring reliable database-to-Kafka event delivery.
- Engineered worker-side retry handling with exponential backoff, lease-based crash recovery, and idempotent execution guards to safely process at-least-once Kafka deliveries.
- Added Docker Compose deployment for multiple scheduler and worker instances, Actuator metrics, failure-scenario documentation, and concurrency tests validating lifecycle transitions and single-claim correctness.

Stock Research Assistant | stockresearchassistant.web.app | FastAPI, LangGraph, LLMs, PostgreSQL, RAG Jul 2026

- Engineered a **multi-agent equity research system** using LangGraph for planning, tool execution, analysis, guardrails, and verification.
- Built a **tool-calling architecture** that selects market, news, financials, price-history, calendar, holders, recommendations, and earnings services by user intent.
- Implemented **PostgreSQL and pgvector-based RAG** for filings, earnings transcripts, and research documents, reducing redundant external API calls.
- Added **verification and source-grounding workflows** to reconcile structured data with document evidence before generating traceable reports.

CricManager | cricmanagernow.web.app | Spring Boot, Flutter, PostgreSQL, Docker, Python Jun 2026

- Built a cricket analytics and simulation platform combining a Spring Boot backend, PostgreSQL persistence, and a Flutter Web frontend for draft-room and tournament workflows.
- Developed a Python ML pipeline using gradient boosting and calibrated logistic regression to estimate ball-by-ball expected runs and win probability from historical cricket data.
- Engineered REST APIs and simulation services to model scores, wickets, match outcomes, and tournament fixtures, then deployed the full stack through Docker and Firebase hosting.

JackDSQL | jackdsq.web.app | Spring Boot, PostgreSQL, Flutter, RabbitMQ, Redis, Docker May 2026

- Built an online SQL practice platform with **270 SQL challenges** and 29 foundational exercises for interactive query execution.
- Designed isolated databases with automated cleanup, query validation, and **safety controls** for user-submitted SQL.
- Integrated **LLM-powered hints and result feedback** for context-aware debugging based on query execution results.